

Dark energy equation of state from spectral geometry on a five-dimensional warped orbifold

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We derive the dark energy equation of state from the spectral action principle applied to a five-dimensional Randall–Sundrum orbifold with internal geometry given by the del Pezzo surface dP_6 (the Fermat cubic). The Gauss–Bonnet Kaluza–Klein reduction yields a geometric coefficient $C_{GB} = 2/3$ (exact), producing a cuscuto-type dark energy sector with superluminal sound speed $c_s \sim 12$ – $15c$. The single remaining free parameter—the Goldberger–Wise scaling dimension $\Delta = 2 + \varepsilon_{GW}$ —is fixed by DESI BAO data at $\varepsilon_{GW} = 0.275^{+0.072}_{-0.106}$, yielding $w_0 = -0.830$ and $w_a = 0$ (exactly). Every other coefficient in the effective action is derived from the spectral triple. The one-loop threshold correction is topological: $\ln(K^2)/\sqrt{K^2 - 1} = \ln 3/\sqrt{2}$, following from the self-intersection number $K^2(dP_6) = 3$. Balanced-metric eigenvalue computations confirm the Kähler–Einstein geometry numerically ($\lambda_1 = 1.461$, 97.4% of the Lichnerowicz bound). The framework provides an explicit UV completion of quadratic quantum gravity in which the R^2 coefficients are geometric consequences, not free parameters.

Recent work on quadratic quantum gravity (QQG) has shown that adding R^2 and $R_{\mu\nu}R^{\mu\nu}$ terms to the Einstein–Hilbert action can eliminate the Big Bang singularity and generate inflation without an inflaton field [1]. These higher-curvature corrections appear with free coefficients that must be determined by observation. We show that the spectral action principle [2, 3], applied to a specific five-dimensional geometry, *derives* the quadratic gravity sector with fixed coefficients and yields a single-parameter cosmological prediction matching current data.

The spectral action principle posits that the bosonic action of a physical theory is $\text{Tr } f(D/\Lambda)$, where D is the Dirac operator of a spectral triple $(\mathcal{A}, \mathcal{H}, D)$ and f is a cutoff function [2]. Applied to the Standard Model spectral triple on a 4D Riemannian manifold, this reproduces the full bosonic Lagrangian of gravity coupled to the Standard Model [4, 5]. We extend this to a 5D Randall–Sundrum (RS) orbifold [6] with internal geometry given by the del Pezzo surface dP_6 , selected by the algebraic chain

$$J_3(\mathbb{O}) \xrightarrow{F_4 \supset E_6} dP_6 \xrightarrow{K^2=3} \text{threshold}. \quad (1)$$

The exceptional Jordan algebra $J_3(\mathbb{O})$ has structure group E_6 [15, 16], which is the root lattice of dP_6 . This fixes the internal geometry without adjustable choices.

Spectral triple and KK reduction.—The five-dimensional geometry is the RS orbifold $M_4 \times S^1/\mathbb{Z}_2$ with warp factor $a(y) = e^{-k|y|}$, where k is the AdS_5 curvature and $y \in [0, y_c]$ is the orbifold coordinate. The spectral action for the 5D Dirac operator produces, via the Seeley–DeWitt expansion through order $a_{7/2}$, the full bosonic action including Gauss–Bonnet (GB) terms. Upon KK reduction to 4D, the GB correction modifies the Israel junction conditions at the IR brane [7], producing an effective kinetic function for the

radion-stabilised scalar ϕ :

$$P(X) = \mu^2 \sqrt{2X} + \epsilon_1 X, \quad (2)$$

where $X = -\frac{1}{2}(\partial\phi)^2$, μ^2 is the brane mass parameter, and $\epsilon_1 = \hat{\alpha} \times C_{GB}$ is the GB correction strength.

The geometric coefficient C_{GB} is determined analytically from the GB-modified Davis junction conditions on the $\mathbb{Z}_2$ orbifold:

$$C_{GB} = \frac{I_4}{I_2} \cdot \frac{2k^2}{(A')^2} \Big|_{\text{brane}} \times f_P = \frac{1}{2} \times 2 \times \frac{2}{3} = \frac{2}{3}, \quad (3)$$

where $I_4/I_2 = 1/2$ is the KK integral ratio, the warp-factor derivative gives a factor of 2, and $f_P = 6/(6+3) = 2/3$ is the P -tensor fraction of gravitational sources at $\mathcal{O}(H^2)$. This has been verified by three independent methods: symbolic tensor computation, direct spectral sum with brane boundary correction (-0.33%), and KK integral evaluation.

The spectral-action coupling $\hat{\alpha}$ is computed from four families of cutoff functions (sharp, exponential, Gaussian, Lorentzian), yielding $\hat{\alpha} = 0.015 \pm 0.002$ after the $d = 5$ Weyl decomposition correction factor $13/22$. Combined with $C_{GB} = 2/3$:

$$\epsilon_1 = \hat{\alpha} \times \frac{2}{3} = 0.010 \pm 0.002. \quad (4)$$

Dark energy from the cuscuto.—The kinetic function (2) with $\epsilon_1 \ll 1$ produces a cuscuto-type dark energy sector [9]. The leading $\sqrt{2X}$ term satisfies $K_{\text{eff}} = P_X + 2XP_{XX} = 0$, the defining property of a cuscuto: zero propagating scalar degrees of freedom. The GB correction $\epsilon_1 X$ breaks the exact cuscuto constraint, producing a sound speed

$$c_s^2 = \frac{C_q}{\epsilon_1}, \quad C_q \equiv \frac{2(1-q_0)}{3|1+q_0|}, \quad (5)$$

where q_0 is the deceleration parameter. For $q_0 \in [-0.53, -0.39]$, this gives $c_s \in [12c, 15c]$ —superluminal but non-pathological. The Gauss–Bonnet correction reintroduces a propagating mode, so this is a genuine front velocity rather than a phase-velocity artefact; causality survives because the RS geometry supplies a preferred frame (the brane rest frame) in which the Babichev–Mukhanov–Vikman closed-signal-curve construction cannot be carried out [10, 11].

The brane mass parameter μ^2 is determined by the Goldberger–Wise (GW) stabilisation mechanism [8]. The GW scalar with scaling dimension $\Delta = 2 + \varepsilon_{\text{GW}}$ satisfies Robin boundary conditions whose eigenvalues determine μ^2 via the spectral action. All ingredients except ε_{GW} are derived from the spectral triple. The algebraic chain is

$$\textcircled{O} \xrightarrow{M_{\text{oct}}} Y_f \xrightarrow{b_{3/2}} \alpha_{\text{UV}} \xrightarrow{\text{Robin}} \mu^2(\varepsilon_{\text{GW}}) \xrightarrow{\text{JC}} w_0, \quad (6)$$

where M_{oct} is the democratic mass matrix from the octonionic mixing (eigenvalues $\{1/2, 1/2, 2\}$, S_3 -symmetric), $b_{3/2}$ is the fermion profile at the one-loop unification scale, α_{UV} is the UV coupling from the spectral action, and the junction conditions (JC) at the IR brane produce the equation of state.

Fitting ε_{GW} to the DESI 2024 BAO data [12] yields $\varepsilon_{\text{GW}} = 0.275^{+0.072}_{-0.106}$ at 1σ , giving:

$$w_0 = -0.830, \quad w_a = 0 \text{ (exactly)}. \quad (7)$$

The $w_a = 0$ prediction is structural: the RS geometry produces a constant equation of state at late times (the warp factor is static after stabilisation). This is testable by DESI Year 5 at 3.8σ against the w_0 – w_a posterior.

Topological threshold.—The one-loop gauge coupling threshold correction from the internal dP_6 geometry involves the analytic torsion of the Kähler–Einstein (KE) metric. For a del Pezzo surface of degree d with canonical class self-intersection $K^2 = d$, the threshold ratio is

$$\frac{a_1}{a_2} = \frac{\ln K^2}{\sqrt{K^2 - 1}} = \frac{\ln 3}{\sqrt{2}} \approx 0.777, \quad (8)$$

which depends only on the topological invariant $K^2 = 3$. The existence of the KE metric is guaranteed by the Tian–Yau theorem [13].

Numerical validation.—We verify the KE geometry using the Donaldson balanced metric algorithm [14] at embedding degrees $k = 8, 12, 15$ on the Fermat cubic $a^3 + b^3 + c^3 + 1 = 0 \subset \mathbb{CP}^2$. At $k = 8$ (109 sections, $(\mathbb{Z}/3)^2$ block decomposition, $\delta G = 5.3 \times 10^{-9}$), the Galerkin eigenvalue computation with 196 basis functions on 600,000 sample points yields $\lambda_1 = 1.461$, recovering 97.4% of the KE Lichnerowicz bound $\lambda_1 \geq 3/2$.

Eigenvector decomposition under $(\mathbb{Z}/3)^2$ reveals that λ_1 and $\lambda_2 = 1.494$ occupy different S_3 representations (standard vs. trivial orbit), confirming their near-degeneracy is accidental. The S_3 splitting $\Delta\lambda_{S_3} =$

$|\lambda_{(1,0)} - \lambda_{(0,1)}|$ shrinks by 31% from $k = 8$ to $k = 15$ (Table I), confirming convergence toward the S_3 -symmetric KE geometry.

Discussion.—The framework presented here provides a concrete UV completion of the quadratic gravity program. Where QQG [1] introduces R^2 corrections with adjustable coefficients, the spectral action on the 5D warped orbifold *derives* these corrections with $C_{\text{GB}} = 2/3$, fixed by the Davis junction conditions. The entire bosonic action—gravity, Standard Model gauge fields, Higgs sector, and dark energy—emerges from a single spectral triple with one undetermined cosmological parameter (ε_{GW}).

The falsifiable predictions are: (i) $w_0 = -0.830$, testable by DESI Y5; (ii) $w_a = 0$ exactly, testable at 3.8σ ; (iii) $c_s \in [12c, 15c]$, constrainable by CMB lensing and large-scale structure; (iv) $\alpha_T = 0$ for gravitational-wave speed (GW170817-compatible). The internal geometry prediction— $K^2(\text{dP}_6) = 3$ selecting the threshold via (8)—is topological and therefore exact.

Full derivations, the 170-page monograph including the fermion sector (octonionic Yukawa couplings, neutrino masses, leptogenesis), and all numerical computations are available in the companion paper [17].

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TABLE I. S_3 splitting convergence across balanced-metric embedding degrees. $\lambda_{(q_1, q_2)}$ denotes the first eigenvalue in the (q_1, q_2) charge sector.

k	N_{sec}	$\lambda_{(1,0)}$	$\lambda_{(0,1)}$	$\Delta\lambda_{S_3}$
8	109	1.461	1.741	0.279
12	235	1.482	1.695	0.214
15	361	1.483	1.675	0.193

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